

University Interscholastic League
2023 – 2024 Elementary Number Sense Test C

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $13 + 51 =$ _____</p> <p>(2) $28 + 86 =$ _____</p> <p>(3) $96 - 44 =$ _____</p> <p>(4) $510 \div 5 =$ _____</p> <p>(5) $19 \times 7 =$ _____</p> <p>(6) $703 - 54 =$ _____</p> <p>(7) $9 + 12 + 15 =$ _____</p> <p>(8) $11 \times 45 =$ _____</p> <p>(9) $360 \div 4 =$ _____</p> <p>*(10) $2023 + 599 =$ _____</p> <p>(11) $18 \times 25 =$ _____</p> <p>(12) Which digit is in the ten-thousandths place in 49317.20568? _____</p> <p>(13) $44 \times 15 =$ _____</p> <p>(14) 83670.2874 rounded to the tenths place is _____ decimal</p> <p>(15) What is the remainder for $70226 \div 9$? _____</p> <p>(16) The number of even whole numbers between 104 and 118 is _____</p> <p>(17) $2 \times 10^{-2} + 9 \times 10^{-3} =$ _____ (decimal)</p> <p>(18) $32 \times 12 - 12 \times 28 =$ _____</p> <p>(19) MMDIX = _____ (Arabic Numeral)</p> | <p>*(20) $625 \times 120 =$ _____</p> <p>(21) $9 - 6 \div 3 =$ _____</p> <p>(22) $13 + 17 + 21 + 25 =$ _____</p> <p>(23) 13 weeks = _____ days</p> <p>(24) Which is larger: 0.66 or $\frac{2}{3}$? _____</p> <p>(25) $\frac{17}{24} - \frac{5}{24} =$ _____</p> <p>(26) $18 \times 22 =$ _____</p> <p>(27) 72% = _____ common fraction</p> <p>(28) If 48 ♥ costs 60¢, then 36 ♥ cost _____ ¢</p> <p>(29) $48 \div \frac{3}{8} =$ _____</p> <p>*(30) $120 \times 667 =$ _____</p> <p>(31) 60 nickels = _____ quarters</p> <p>(32) The sum of the two largest primes less than 40 is _____</p> <p>(33) 40 ounces = _____ cups</p> <p>(34) $825 \div 25 =$ _____</p> <p>(35) $41\frac{2}{3}\%$ = _____ common fraction</p> <p>(36) The LCM of 12 and 24 is _____</p> <p>(37) $91 \times 95 =$ _____</p> |
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- (38) $\frac{11}{24} + \frac{7}{24} - \frac{3}{24} =$ _____
- (39) The ratio, in cents, of 6 dimes to 3 quarters is _____
- *(40) $232 \times 225 =$ _____
- (41) $5^3 =$ _____
- (42) $23^2 =$ _____
- (43) The volume of a cube with edge 5 is _____
- (44) The perimeter of an isosceles triangle with congruent sides 18 and other side 24 is _____
- (45) $4^3 - 3^3 =$ _____
- (46) $4\frac{1}{2} \times 2\frac{1}{2} =$ _____ (mixed number)
- (47) What is the volume of a rectangular box that is 2.25-ft by 4-ft by 12-ft? _____ cubic feet
- (48) $800 \times 12.5\% =$ _____
- (49) If $x = \frac{3}{4}$, then $24 - 4x =$ _____
- *(50) $10 \times 2024 + 2025 =$ _____
- (51) What is the number, **k**, in the sequence: 0, 1, 1, 2, 3, 5, **k**, 13, 21, . . . ? _____
- (52) $8\frac{1}{5} - 5\frac{7}{10} =$ _____ (mixed number)
- (53) If the diameter of a circle is 44, then the area of the circle is _____ π
- (54) If a rectangle with area 28 has a length that measures 14, then its perimeter is _____
- (55) 48 (base 10) = _____ (base 5)
- (56) The number of days in March is _____
- (57) What is the area of a triangle with altitude $4\frac{1}{3}$ and base 18? _____
- (58) How many elements are in the power set for {T, E, N}? _____
- (59) If four times a whole number, added to six, is thirty-eight, what is the number? _____
- *(60) 12 miles = _____ yards
- (61) 1001 (base 2) = _____ (base 8)
- (62) $18 - (-2^4) \times (2) =$ _____
- (63) What is the probability a red marble is randomly picked from a bag of 22 blue and 8 red marbles? _____
- (64) 4.75 gallons = _____ pints
- (65) $69^2 =$ _____
- (66) $10^5 \div 8$ has remainder of _____
- (67) What is the perimeter of a rhombus with side 6.125? _____
- (68) If $6x - 14 > 46$, then $x >$ _____
- (69) $2\frac{4}{35} - \frac{7}{5} =$ _____
- *(70) $\sqrt{115600} =$ _____
- (71) The multiplicative inverse of $4\frac{5}{8}$ is _____
- (72) $21^2 - 29^2 =$ _____
- (73) If $6\frac{1}{2}\%$ of 12 is $1\frac{5}{8}\%$ of x , then $x =$ _____
- (74) $1^0 + 2^1 - 3^2 + 4^3 =$ _____
- (75) $12^2 + 36^2 =$ _____
- (76) $(2 + 4 + 6 + \dots + 10)^2 =$ _____
- (77) What is the area of a right triangle with hypotenuse 26 and leg 24? _____
- (78) $\sqrt{324} \times \sqrt{484} =$ _____
- (79) $39 + 39^2 =$ _____
- *(80) $16^4 =$ _____

2023 – 2024 University Interscholastic League Elementary Number Sense Test C – Key

(1) 64	*(20) 71250 – 78750	(38) $\frac{5}{8}$; .625	(59) 8
(2) 114	(21) 7		*(60) 20064 – 22176
(3) 52	(22) 76	(39) $\frac{4}{5}$; .8	(61) 11
(4) 102	(23) 91		(62) 50
(5) 133	(24) $\frac{2}{3}$	*(40) 49590– 54810	(63) $\frac{4}{15}$
(6) 649		(41) 125	(64) 38
(7) 36	(25) $\frac{1}{2}$; .5	(42) 529	(65) 4761
(8) 495	(26) 396	(43) 125	(66) 0
(9) 90		(44) 60	
*(10) 2491 – 2753	(27) $\frac{18}{25}$	(45) 37	(67) 24.5; $24\frac{1}{2}$; $\frac{49}{2}$
(11) 450	(28) 45	(46) $11\frac{1}{4}$	(68) 10
(12) 6	(29) 128	(47) 108	
(13) 660	*(30) 76038– 84042	(48) 100	(69) $\frac{5}{7}$
(14) 83670.3	(31) 12	(49) 21	*(70) 323 – 357
(15) 8	(32) 68	*(50) 21152 – 23378	(71) $\frac{8}{37}$
(16) 6	(33) 5	(51) 8	(72) -400
(17) .029	(34) 33	(52) $2\frac{1}{2}$	(73) 48
(18) 48	(35) $\frac{5}{12}$	(53) 484	(74) 58
(19) 2509	(36) 24	(54) 32	(75) 1440
	(37) 8645	(55) 143	(76) 900
		(56) 31	(77) 120
		(57) 39	(78) 396
		(58) 8	(79) 1560
			*(80) 62260 – 68812

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.